

Aarya Venkat, PhD

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PROFESSIONAL SUMMARY

Computational chemist with over 2.5 years of industry experience in computational protein and drug discovery. Expert in integrating SOTA machine learning with physics-based modeling to accelerate protein discovery and optimization. Proven track record of building automated MLOps pipelines and leading cross-functional projects in fast-paced environments.

EDUCATION

University of Georgia — PhD Computational Chemistry — Dec 2023

Honors: ARCS Foundation Scholar, Future Faculty Fellow, Three Minute Thesis award

University of California, San Diego — MS Chemistry — June 2017

TECHNICAL SKILLS

Programming: Python, Bash, Git, Snowflake, Docker, Kubernetes, Argo Workflows, Nextflow, AWS/GCP

Bioinformatics: Metagenomic Sourcing, Sequence analysis, Intron/Exon analysis, NGS, Clustering

Cheminformatics: RDKit, QSAR, xTB, CREST, molecular docking, virtual screening

Computational Chemistry: Molecular Dynamics, DFT, MDAnalysis, Amber, GROMACS, Desmond, PyMol, Schrodinger Maestro, forcefield parameterization, BioEmu, protein-ligand interaction analysis

Protein Design: Inverse Folding (ProteinMPNN, LigandMPNN), EVCouplings, Mutational inference (ThermoMPNN, ESM2), co-folding, Predictive modeling, de novo protein design (RFDiffusion),

AI/ML: PyTorch, Explainable AI (SHAP), linear/logistic regression, Protein Language Models (ESM2), fine-tuning (PEFT, LoRA), Dataset preparation, training ML models for protein optimization

Research & Soft Skills: Technical Writing, Data Visualization and Presentation, Public Speaking

PROFESSIONAL EXPERIENCE

Ginkgo Bioworks, Inc. — Scientist III

Dec 2023 – Present — Boston, MA

- Designed T7 RNA Polymerase as Ginkgo Bioworks' first **commercial enzyme product** with 99.6% reduction (over wt T7) of dsRNA, a toxic byproduct with immunogenic effects.
- Lead over 35 protein engineering campaigns utilizing AI/ML, molecular modeling, and evolutionary models to enhance specificity, thermostability, and activity for high-budget contracts.
- Optimized enzymes to reach 45x higher activity over wt in three rounds, with tiny library sizes of 200-350 variants per round.
- Utilized and combined metagenomic sourcing with structure-prediction tools (ESMFold, Chai, Boltz) to achieve protein discovery objectives, identifying proteins with 10x higher activity, thermostability at 90C, halotolerant (stable under 1M salt), along with identification of enzymes that perform novel chemistry, previously unseen in literature.
- Serve as Subject Matter Expert and Tech Liaison for AI and computational biology, leading customer-facing meetings to communicate research benefits, risks, and strategic roadmaps.
- Architected **Kubernetes** pipelines using **Argo** and **Docker** for atomic workflows like **RFDiffusion-AA**, proteinMPNN, as well as larger meta-workflows created to automate protein design.
- Developed **PyTorch** deep learning models fine-tuned via **LoRA** from ESM2 to build family-specific enzyme models for Ginkgo Bioworks' 'enzyme custom plates' service.

University of Georgia — Graduate Research Assistant

Aug 2017 – Dec 2023 — Athens, GA

- Authored 17+ peer-reviewed manuscripts, contributing to writing, experimental design, data visualization, and successful funding acquisitions.
- Conducted advanced evolutionary analyses including molecular dynamics, phylogeny, and quantum chemistry to study enzyme function.
- Managed lab operations like data storage infrastructure, procurement, and protocol documentation.

PRODUCTS

T7 ClearCirc RNA Polymerase (Inventor): Directed the protein engineering of specific RNAP mutations reducing toxic dsRNA byproduct by 99.6%, significantly reducing immunogenicity and streamlining downstream purification for in vitro and in vivo applications. Technology serves as the core of a commercial drop-in replacement kit for high-fidelity circular RNA synthesis.

SELECTED PUBLICATIONS

Total: 23 — Citations: 444 — H-index: 12 — co-first author[†]

Venkat, A. (2026). GapClean, a tool for clarifying protein sequence alignments. *In Prep.*

Fang, R. et al. (2026). Glydentify: An explainable deep learning platform for glycosyltransferase donor substrate prediction. *bioRxiv*.

Zhang, J. et al. (2025). ICEPIC: A Toolkit to Discover Ice Binding Proteins from Sequence. *bioRxiv*.

Venkat, A. et al. (2023). Mechanistic and evolutionary insights into isoform-specific ‘supercharging’ in DCLK family kinases. *eLife*.

Zhou, Z. et al. (2023). xAI meets biology: A comprehensive review of explainable ai in bioinformatics applications. *arXiv*.

Meng, Y. et al. (2023). Phosphorylation-dependent pseudokinase domain dimerization drives full-length MLKL oligomerization. *Nature Communications*.

Aceil, J.[†], **Venkat, A.**[†], et al. (2023). Prevalence and homology of the pneumococcal serine-rich repeat protein at the global scale. *Microbiology Spectrum*.

Venkat, A., et al., (2022). Modularity of the hydrophobic core and evolution of functional diversity in fold A glycosyltransferases. *Journal of Biological Chemistry*.

Taujale, R., **Venkat, A.**, et al. (2020) Deep evolutionary analysis reveals the design principles of fold A glycosyltransferases. *eLife*.

PATENTS

Venkat, A., Castro, Y., and Manning, T. Using Computational QSAR Methods to Propose a New Group of Antibiotics for Dental Applications, U.S. Provisional Patent Appl. 62/359,638, July 22, 2016.

AWARDS

2023 — Eriksson Lecture Scholarship

2023 — ASBMB Conference Travel Award

2021 — ARCS Foundation Scholar

2021 — Three Minute Thesis Award

2021 & 2022 — Glycobiology Poster Award & Travel Award